# STANDATA variance 23-BCV-009

Building

## CSA B52:23 Mechanical refrigeration code

Date Issued: October 2024



**Purpose**

Allows the acceptance of CSA B52:23, “Mechanical refrigeration code,” under the National Building Code – 2023 Alberta Edition (NBC(AE)).

**Discussion**

On January 1, 2025, new standards for manufacturing with refrigerants will go into effect.

These standards include prohibiting the manufacture of products containing certain hydrofluorocarbons (HFCs) by January 1, 2025. HFCs are synthetic organic compounds used in air conditioning and as refrigerants.

To conform to these rule changes, the 2023 edition of the CSA B52 code introduced the A2L and B2L refrigerant groups as more environmentally friendly refrigerants. Equipment manufacturers have also begun to adapt to the rule changes by re-tooling their factories to manufacture new equipment to allow for the use of these A2L and B2L refrigerant groups. Additionally, there is already equipment on the market which use the new refrigerants.

The NBC(AE), however, currently references the 2018 edition of the CSA B52 Code, as per Table 1.3.1.2. of Division B. To align with the HFC prohibition and industry’s change in supply of HVAC equipment across Canada, it is necessary for the NBC(AE) to recognize the CSA B52:23 Code.

**Code References**

**Sentence 6.2.1.5.(1)**

**6.2.1.5. Installation Standards**
1) Except as provided in Articles 6.9.4.2. and 6.3.1.4., the installation of heating and air-conditioning equipment, including mechanical refrigeration equipment, and including provisions for mounting, clearances and air supply, shall conform to the requirements of
- a) the Pressure Equipment Safety Regulation made pursuant to the Safety Codes Act,
- b) CSA B52, “Mechanical refrigeration code,”
- c) CSA B139 Series, “Installation code for oil-burning equipment,”
- d) the Gas Code Regulation made pursuant to the Safety Codes Act,
- e) CSA B365, “Installation Code for Solid-Fuel-Burning Appliances and Equipment,” or
- f) the Electrical Code Regulation made pursuant to the Safety Codes Act.



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**Sentence 9.33.5.2.(1)**

**9.33.5.2. Installation Standards**
1) Except as provided in Articles 9.33.5.3. and 9.33.5.4., the installation of heating and air-conditioning equipment, including mechanical refrigeration equipment, and including provisions for mounting, clearances and air supply, shall conform to
a) the Pressure Equipment Safety Regulation made pursuant to the Safety Codes Act,
b) CSA B52, "Mechanical refrigeration code,"
c) CSA B139 Series, "Installation code for oil-burning equipment,"
d) the Gas Code Regulation made pursuant to the Safety Codes Act,
e) the Electrical Code Regulation made pursuant to the Safety Codes Act, or
f) CAN/CSA-C448 Series, "Design and installation of earth energy systems."
(See also Sentence 9.33.5.3.(1).)

**Division B, Table 1.3.1.2.**

| Issuing Agency | Document Number | Title of Document | Code Reference |
|---|---|---|---|
| CSA | B52:18 | Mechanical refrigeration code | 6.2.1.5.(1)<br>9.33.5.2.(1) |

**Application**

This variance applies to the installation of equipment required to conform to CSA B52, as per Sentences 6.2.1.5.(1) and 9.33.5.2.(1) of the NBC(AE).

**Variance**

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the NBC(AE).

1. Conformance with CSA B52:23, "Mechanical refrigeration code," is considered to comply with the installation standard requirements of the NBC(AE).

This VARIANCE is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

Issued by the Provincial Building Administrator

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